# THE BLOW-UP SQUARE OF A FINITE TYPE QUASI-COHERENT IDEAL GIVES A COHOMOLOGICAL DESCENT DISTINGUISHED TRIANGLE FOR TORSION COEFFICIENT COMPLEXES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the proof is the author's complete argument: the choice of the complexes representing the three derived pushforwards, the identification by proper base change, the reduction to a stalk statement via the triangle attached to a short exact sequence of complexes, and the two-case stalk verification are all written out.

*Editorial note.* The label lemma-blow-up-square-cohomological-descent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the four cited inputs are kept as external pointers with their canonical Stacks tags.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- The blow-up square of a closed subscheme cut out by a quasi-coherent ideal of finite type. Same chapter, the blow-up square with E = Z x_X X' and the equalities c = i o pi = b o j. Supplies the geometric set-up of the statement and the notation for the four morphisms.
- Proper base change and its pushforward form for the blow-up. Same chapter, the proper base change theorem and its consequence identifying pi_*(I^bullet|_E) with (b_*I^bullet)|_Z. Supplies the identification of the complex representing R c_*(K|_E) used to write the map of complexes.
- Exactness of the pushforward along a finite morphism. Same chapter, the vanishing of the higher direct images of a finite morphism, used for the closed immersion i. Supplies that i_*(F^bullet|_Z) represents R i_*(K|_Z).

*Source reference.* Original section 014Z: sources/stacks-chow/tags.txt, source lines 1490–1490. Prerequisite pointer invoked at proof line(s) 23368; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 09V5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18708–18708. Prerequisite pointer invoked at proof

*Date*: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

line(s) 23348; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 0A3U: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 18282–18282. Prerequisite pointer invoked at proof line(s) 23347; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Prerequisite pointer invoked at proof line(s) 23339; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $X$ be a scheme and let $Z \subset X$ be a closed subscheme cut out by a quasi-coherent ideal of finite type. Consider the corresponding blow up square*

$$\begin{array}{ccc} E & \xrightarrow{j} & X' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle b} \\ Z & \xrightarrow{i} & X \end{array}$$

*For $K \in D^+(X_{étale})$ with torsion cohomology sheaves we have a distinguished triangle*

$$K \to Ri_*(K|_Z) \oplus Rb_*(K|_{X'}) \to Rc_*(K|_E) \to K[1]$$

*in $D(X_{étale})$ where $c = i \circ \pi = b \circ j$.*

*Proof.* The notation $K|_{X'}$ stands for $b_{small}^{-1}K$. Choose a bounded below complex $\mathcal{F}^\bullet$ of abelian sheaves representing $K$. Observe that $i_*(\mathcal{F}^\bullet|_Z)$ represents $Ri_*(K|_Z)$ because $i_*$ is exact (Proposition 03QP). Choose a quasi-isomorphism $b_{small}^{-1}\mathcal{F}^\bullet \to \mathcal{I}^\bullet$ where $\mathcal{I}^\bullet$ is a bounded below complex of injective abelian sheaves on $X'_{étale}$. This map is adjoint to a map $\mathcal{F}^\bullet \to b_*(\mathcal{I}^\bullet)$ and $b_*(\mathcal{I}^\bullet)$ represents $Rb_*(K|_{X'})$. We have $\pi_*(\mathcal{I}^\bullet|_E) = (b_*\mathcal{I}^\bullet)|_Z$ by Lemma 0A3U and by Lemma 09V5 this complex represents $R\pi_*(K|_E)$. Hence the map

$$Ri_*(K|_Z) \oplus Rb_*(K|_{X'}) \to Rc_*(K|_E)$$

is represented by the surjective map of bounded below complexes

$$i_*(\mathcal{F}^\bullet|_Z) \oplus b_*(\mathcal{I}^\bullet) \to i_*\left(b_*(\mathcal{I}^\bullet)|_Z\right)$$

To get our distinguished triangle it suffices to show that the canonical map $\mathcal{F}^\bullet \to i_*(\mathcal{F}^\bullet|_Z) \oplus b_*(\mathcal{I}^\bullet)$ maps quasi-isomorphically onto the kernel of the map of complexes displayed above (namely a short exact sequence of complexes determines a distinguished triangle in the derived category, see Derived Categories, Section 014Z). We may check this on stalks at a geometric point $\overline{x}$ of $X$. If $\overline{x}$ is not in $Z$, then $X' \to X$ is an isomorphism over an open neighbourhood of $\overline{x}$. Thus, if $\overline{x}'$ denotes the corresponding geometric point of $X'$ in this case, then we have to show that

$$\mathcal{F}^\bullet_{\overline{x}} \to \mathcal{I}^\bullet_{\overline{x}'}$$

is a quasi-isomorphism. This is true by our choice of $\mathcal{I}^\bullet$. If $\overline{x}$ is in $Z$, then $b_(\mathcal{I}^\bullet)_{\overline{x}} \to i_*\left(b_*(\mathcal{I}^\bullet)|_Z\right)_{\overline{x}}$ is an isomorphism of complexes of abelian groups. Hence the kernel is equal to $i_*(\mathcal{F}^\bullet|_Z)_{\overline{x}} = \mathcal{F}^\bullet_{\overline{x}}$ as desired. □